Day-25 29th Dec

RIP & EIGRP

RIP & EIGRP aren't on official exam topics list, but we may still get some questions

RIP (Routing Information Protocol)

- It's industry standard, not CISCO proprietary.
- Distance vector IGP (uses routing-by-rumor logic) to learn/share routes
- Uses hop count as it's metric. (One router = one hop (bandwidth is irrelevant))
- The maximum hop count is 15 (anything more than that is considered unreachable)
- Has three versions:
  - RIPv1 & RIPv2, used for IPv4
  - RIPnG (RIP Next Gen) used for IPv6

- Uses two message types.
  - Request: To ask RIP-enabled neighbor routers to send their routing table.
  - Response: To send the local router's routing table to neighboring routers.
- By default RIP-enabled routers will share their routing table every 30 sec.

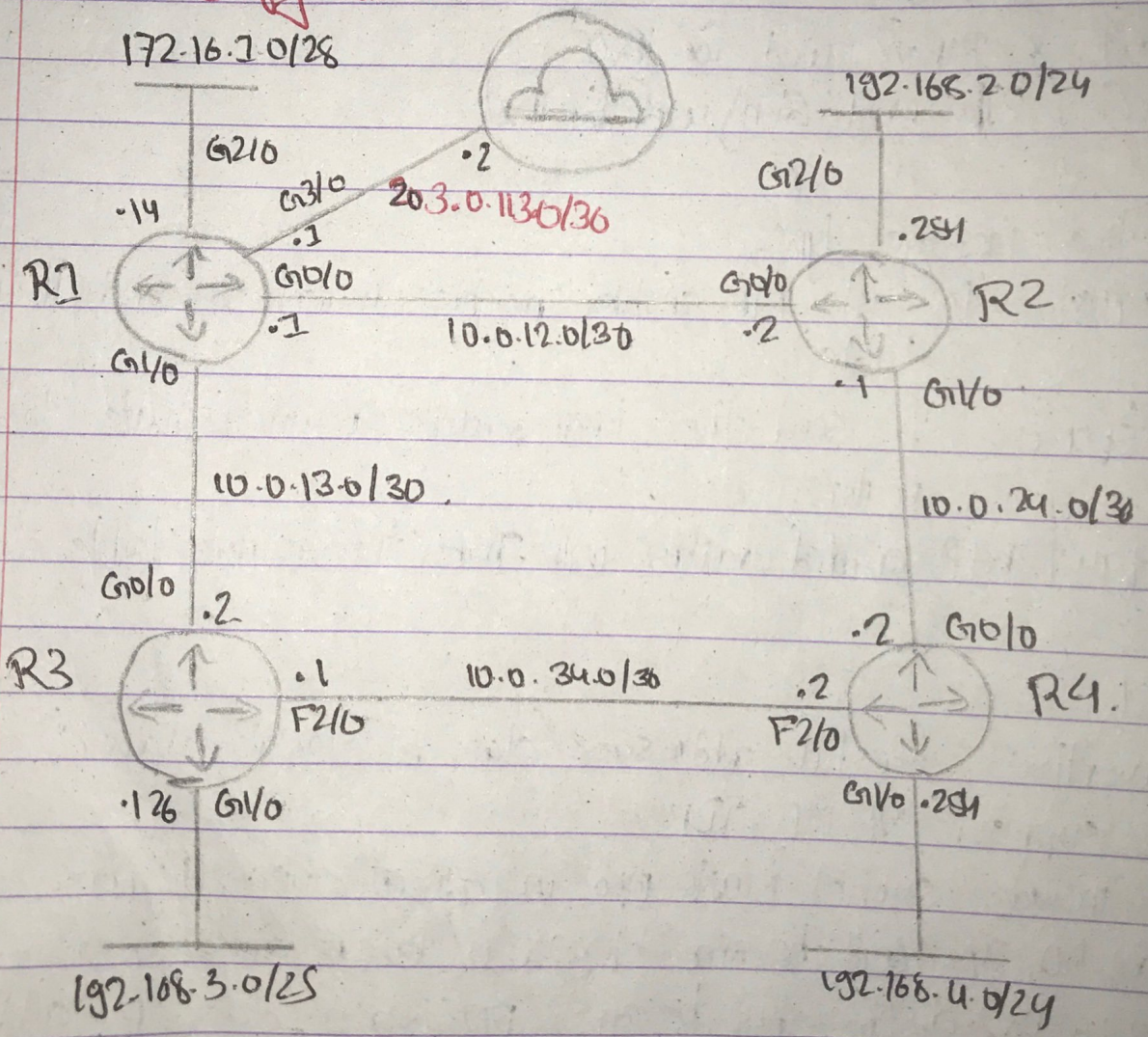
RIPv1:

- only advertises classful addresses (class A, class B, class C)
- doesn't support VLSM, CIDR
- doesn't include subnet mask info in advertisements (Response message)
  - 10.1.1.0/24 will become 10.0.0.0 (class A address, so assumed to be /8)
  - 172.16.192.0/18 will become 172.16.0.0 (class B add, so assumed to be /16)
  - 192.168.1.4/30 will become 192.168.1.0 (class C add, so assumed to be /24)
- messages are broadcast to 255.255.255.255

- RIPv2
  - supports VLSM, CIDR.
  - includes subnet mask into in advertisements.
  - mssg are multicast to 224.0.0.9

- Broadcast mssgs are delivered to all devices on local network. Multicast mssgs are delivered only to devices that have joined that specific multicast group

RIP Configuration.



```
R1(config) # router rip                    (enables RIP on router)
R1(config-router) # version 2
R1 (config-router) # no auto-summary
R1 (config-router) # network 10.0.0.0
R1 (config-router) # network 172.16.0.0
```

- The RIP 'network' command is classful, it will automatically convert to classful networks.
- For examples, even if we enter command network 10.0.12.0, it will be converted to network 10.0.0.0 (a class A network)
- There is no need to enter network mask.
- The network commands tells the router to:
  - look at int with an IP add. that is in the specified range
  - active RIP on the int. that fall on the range
  - form adjacencies with connected RIP neighbors
  - advertise network prefix of interface (NOT prefix on network command)

- The OSPF & EIGRP command operate in the same way.
- Bco3 network command is classful 10.0.0.0 is assumed to be 10.0.0.0/8.
- R1 will look for any int with an IP add that matches 10.0.0.0/8 (because its a /8, it only need to match first /8 bits)
- 10.0.12.1 & 10.0.13.1 both match, so RIP is activated on G0/0 & G1/0
- R1 forms adjacencies with neighbor R2 & R3.
- R1 advertises 10.0.12.0/30 and 10.0.13.0/30 (NOT 10.0.0.0/8) to its RIP neighbors.

Scanned with CamScanner

The Passive Interface Command.

- There are no RIP neighbors connected to G2/0, R1 will continuously send RIP advertisement & out of G2/0. This is unnecessary traffic, so G2/0 should be configured as passive interface.

For this use the command from Router-config mode

R1(config-router)# passive-interface g2/0.

- After using this command router will stop sending RIP advertisements to G2/0, but will advertise the network G2/0/0 is connected to (172.16.1.0/28) to RIP neighbors (R2, R3)

Advertise a default route to eig RIP.

So, in our network only one router is connected to internet All device shall have access to internet. So, how do we configure for this?

R1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.2
R1(router config-router)# default-information originate.

Useful Commands

- show ip protocols
- no auto-summary (For RIP)

By default RIP will insert upto 4 paths to same destination but this can be changed using below command

R1(config-router)# maximum-paths ?
<1-32> No. of paths
R1(config-router)# maximum-paths 8

Scanned with CamScanner

Wildcard masks

In EIGRP, OSPF etc. we use something called wildcard masks to spe when using network command as

R1(conf-router)# network 172.16.1.0 0.0.0.15

But we have 172.16.1.0/28 network and it shall be .255.255.255.239 right?

↳ Yes, but we only if its subnet-mask. But we use wildcard-mask here, which is just inverse of subnet-mask

• Convert 0's of subnet-mask to 1 & vice-versa. We will get wildcard mask.

11111111. 11111111. 11111111. 00000000 } Subnet mask /24
255 . 255 . 255 . 0

↓

00000000.00000000.00000000. 11111111 } Wildcard Mask. /24
0 . 0 . 0 . 255

& 255.255.0.0 /16 } Subnet Mask
0.0.255.255 /16 } Wildcard Mask.

255.0.0.0 /8 } Subnet Mask.
0.255.255.255 /8 } Wildcard Mask

In our R1's G2/0, it has prefix of /28.

11111111. 11111111. 11111111. 11110000 } Subnet Mask
255. 255. 255. 240

↓

00000000. 00000000. 00000000. 00001111 } Wildcard Mask.
0 . 0 . 0 . 15

Subnet - 255.255.255.128
Wildcard - 0. 0. 0. 127

Subnet - 255.252.0.0
Wildcard - 0.3.255.255

Trick is to subtract Subnet each octet from 255.

NOTE:
'0' in the wildcard mask = must match
'1' in the wildcard mask = don't have to match

R1 G2/0 IP add.

10101100. 00010000. 0000 0001 0000 1110
172 . 16 . 1 . 14.

EIGRP network command

10101100. 00010000 . 00000001 . 00000000
172 . 16 . 1 . 0

00000000. 00000000. 00000000. 00001111
0 . 0 . 0 . 15

So, first 28-bits must match as wild-card mask has 0s in first 28-bits. They do match.
So, EIGRP will be activated on interface.
Remember 0's shall be continuous 00010 means first 3 bits shall match.

Scanned with CamScanner

R1 G2/0 IP add.

10101100. 00010000. 00000001. 00001110
172. 16. 1. 14.

EIGRP network command:

10101100. 00010000. 00000001. 00000000
172. 16. 1. 0

00000000. 00000000. 00000000. 00000111
0. 0. 0. 7

Here the 29th bit ~~should~~ be didn't match even Wildard mask states it shall be matched. So, the interface won't go in EIGRP mode.

Day-25

& RIP & EIGRP (contd---)

EIGRP.

- Enhanced Interior Gateway Routing Protocol
- Was Cisco Proprietary, but Cisco has now published its openly so other vendors can implement it to their equipment.

- Considered an 'advanced/hybrid' distance routing protocol.
- Much faster than RIP in reacting to changes in the network.
- Doesn't have 15 'hop-count' limit of RIP.
- Sends messages using multicast address 224.0.0.10.
- Is the only IGP that can perform unequal-cost load-balancing (by default it perform ECMP load-balancing over 4 paths like RIP).

EIGRP configuration.

We can have both RIP & EIGRP running at the same time but it'll just be waste of resource. So, usually we'll have only one IGP running a router.

To setup/configure EIGRP use following commands:

i) R1(config)# router eigrp 1

i.e router eigrp AS_number. (Autonomous system no)

§ The AS no. must match between routers, or they'll not form an adjacency & share route info §.

ii) R1(config-router)# no auto-summary.

It was fn same as RIP, advertises classful networks instead of actual network prefix configured on interfaces.

iii) R1(config-router)# passive-interface g2/0.

It also has passive-int command same as RIP.

Scanned with CamScanner

iii) R1(config-router) # network 10.0.0.0 (a)
R1 (config-router) # network 172.16.1.0 0.0.0.15. (b)

We can use network command with or without mask. However, network command will assume a classful add. if we don't specify the mask.

EIGRP uses a wildcard mask instead of regular subnet mask. So, if that 0.0.0.15 is /28 network.

show ip protocols.

In EIGRP & OSPF, the router has a unique ID which identifies it within the AS

Router ID order of priority:

1) Manual configuration
2) Highest IP add on a ~~loop~~ loopback interface.
3) Highest IP address on a physical interface.

Manual configuration of router-id on EIGRP.

R1(config-router)# eigrp router-id ?
eigrp router-id 1.1.1.1

do show ip eigrp protocol